RUSSIAN-AFRICAN RAW MATERIALS DIALOGUE

The First Higher Technical University in Russia
EMPRESS CATHERINE II SAINT PETERSBURG MINING UNIVERSITY

unesco Centre Under the auspices of UNESCO

International Competence Centre for Mining-Engineering Education under the auspices of UNESCO

НЕДРА

unesco

RUSSIAN-AFRICAN FORUM OF YOUNG SCIENTISTS:
"FUTURE ENGINEERS OF THE WORLD – THE FOUNDATION OF SUSTAINABLE DEVELOPMENT"

Securing the Digital Mine: A Metaheuristic Optimized Deep Learning Framework for Intrusion Detection in IoT Enabled Mineral Resource Operations

•

Nomination: Track 3, "Smart Subsoil": Digital Transformation and Automation in the Mineral Resources Complex

Saint Petersburg
2026

Team presentation

- Team Leader: **John Okyere,**
UEW, BSc. Computer Science. Specializes in metaheuristic optimization, deep learning for intrusion detection, and blockchain security.
- **Ezekiel Baah, Ghana,UEW**, Machine Learning Engineer and Data Scientist
- **Clement Baffour, Ghana, UEW**, Software Engineer and Security Expert
- **George Arhin-Bonnah, Ghana, UEW**, Skilled in Cloud Computing and Backend Engineering
- **Paa Annobil Parker, Ghana, UEW**, Skilled in IOT and SCADA Technologies

Why this team: Combined expertise in AI/ML, cybersecurity, and network intrusion detection. Team Leader has trained 600+ developers across Ghana and internationally, won international hackathons including 1st place at Safe Agentathon 2025, and has three journal manuscripts under peer review in IEEE Access and Oxford financial journals.

Published research on metaheuristic optimized CNN-LSTM for network intrusion detection. Active ICP Ambassador and co-founder of Kayaba Labs, Ghana.

Relevance of the study

African and Russian mining operations are digitalizing rapidly, yet cybersecurity for industrial IoT and SCADA environments remains critically underdeveloped.

• Problem: African mining facilities (e.g., Gold Fields Tarkwa, Ghana) deploy IoT sensors, SCADA, and digital twins. These OT environments are now internet-exposed, creating attack surfaces existing IDS tools cannot address.
• Why now: OT cyberattacks are rising (African Mining Market, 2024). Most IDS tools use IT benchmarks (NSL-KDD) that miss industrial protocol patterns and SCADA polling signatures.
• Key fact: ICS intrusions can halt production or trigger unsafe conditions endangering workers (Kheddar, Himeur & Awad, 2023).
• Ghana: Mining digitalization is accelerating under the Minerals Commission of Ghana, making OT cybersecurity urgently needed.

RUSSIAN-AFRICAN RAW MATERIALS DIALOGUE

Analysis of existing solutions

• Literature review of deep transfer learning for ICS/IIoT intrusion detection (Kheddar, Himeur & Awad, 2023) confirms most IDS approaches are validated on IT datasets and not adapted for OT environments.

• Existing solutions: Signature-based IDS (Snort, Suricata), anomaly-based ML classifiers on NSL-KDD/CICIDS, and emerging deep learning approaches (CNN, LSTM, GRU) for generic network traffic. SCADA-specific datasets (SWaT, BATADAL) exist but are rarely used in resource-constrained African deployment contexts.

• **Key gap: No existing solution combines metaheuristic feature selection with CNN-LSTM specifically adapted and validated for African mining OT/IIoT traffic with edge-deployment constraints in mind.**

Parameter	BWOA + CNN-LSTM (proposed)	Generic ML IDS (NSL-KDD trained)	Signature-based IDS (Snort/Suricata)
Price	Low	Low	High
Efficiency	High	High	High
Reliability	High	Low	Low
Eco-friendliness	High	High	Low

Conclusion: *Conclusion: Our framework is the only proposal combining metaheuristic optimized feature selection with a CNN-LSTM classifier explicitly designed for OT/IIoT mining traffic in resource-constrained African deployment contexts.*

Solution Development/Concept

RUSSIAN-AFRICAN RAW MATERIALS DIALOGUE

Novelty: First systematic framework for adapting a BWOA optimized CNN-LSTM intrusion detection pipeline from IT benchmark traffic (NSL-KDD) to mining OT and IIoT environments, with explicit attention to African infrastructure constraints.

Principle: Binary Whale Optimization Algorithm performs intelligent feature selection on raw network traffic, reducing dimensionality and computational load. The selected features feed a CNN-LSTM classifier that captures both spatial packet-level patterns (CNN) and temporal sequence dynamics (LSTM) characteristic of SCADA and IoT polling traffic.

Raw OT/IoT network traffic from mining SCADA, sensors, and cloud connected systems

CICFlowMeter feature extraction + BWOA feature selection (reduces from 80+ to optimal feature subset)

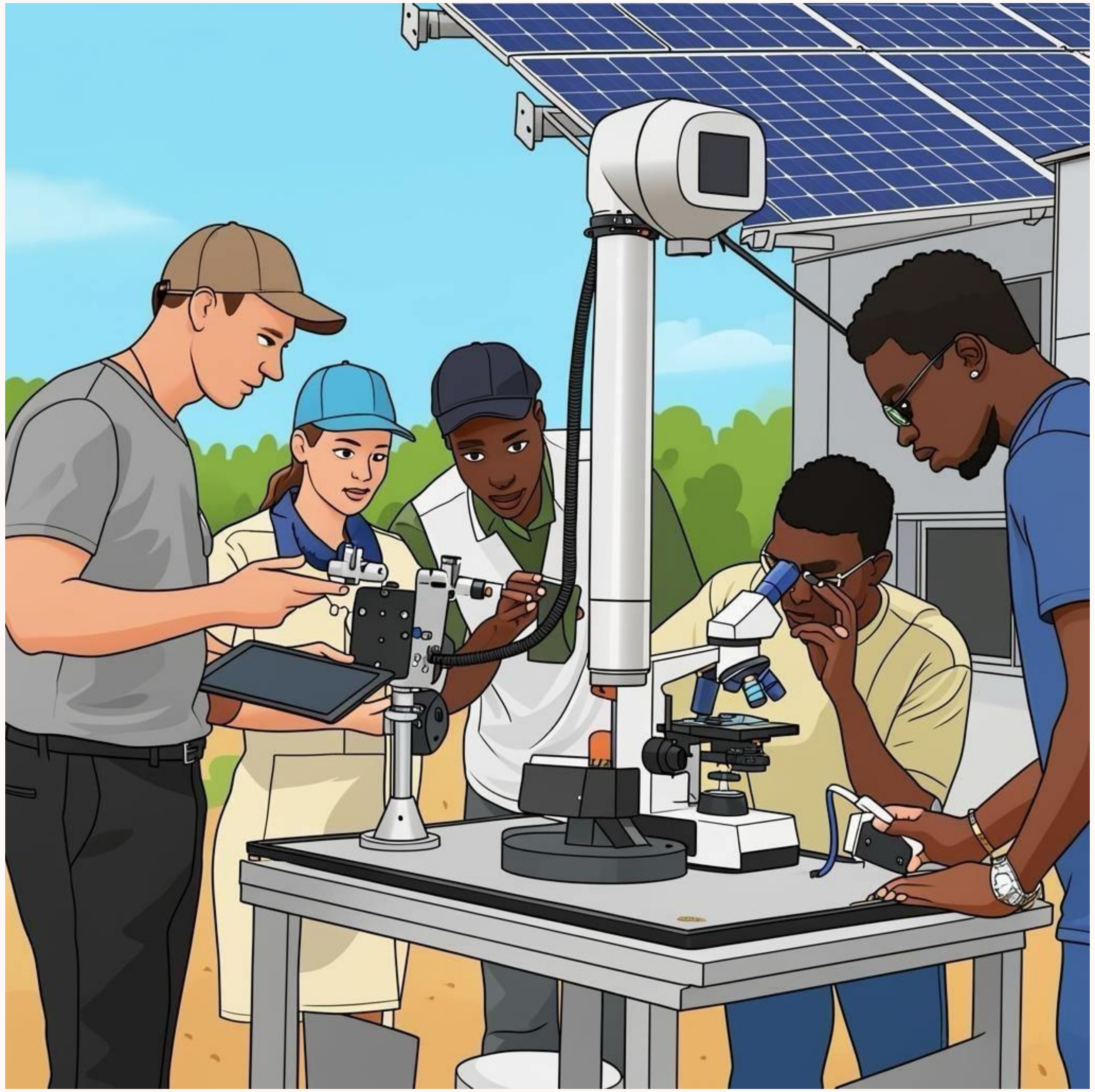
CNN-LSTM classifier trained on OT-specific feature profiles, validated against IIoT intrusion benchmarks

Real-time intrusion alert with attack class label, optimized for edge deployment in remote mining sites

Important: *The Whale Optimization Algorithm mimics humpback whale bubble-net hunting behavior to search for the most informative feature subset, eliminating noise and reducing model size for edge deployment.*

RUSSIAN-AFRICAN RAW MATERIALS DIALOGUE

Technical implementation



Technical approach: BWOA + CNN-LSTM pipeline adapted from NSL-KDD validated baseline toward OT/IIoT mining traffic.

- **Phase 1 (Data):** Collaborative OT traffic capture at mining pilot sites using AWS EC2 + CICFlowMeter instrumented architecture. Target: labeled attack and benign traffic for SCADA protocols (**Modbus, DNP3, OPC-UA**) and **IoT sensor streams**.
- **Phase 2 (Model):** Retrain BWOA feature selector on OT feature space. Retrain CNN-LSTM on collected and augmented IIoT traffic. Validate against SWaT and BATADAL industrial benchmarks. Baseline: CNN-LSTM accuracy of 67.3% on NSL-KDD; target improvement via OT-specific feature engineering.
- **Phase 3 (Deployment):** Evaluate model latency and memory footprint for Raspberry Pi / edge hardware. Target: sub-100ms detection latency on 1GB RAM edge device, compatible with remote mining site connectivity constraints.

Sustainable development

Sustainable Development Goals (SDGs)

This project directly supports three UN Sustainable Development Goals, aligned with both African and Russian industrial development priorities.

Visualization: *UN SDG icons, effect diagrams*

Economic effect

Accessibility of the solution for the local population, stimulating the economy

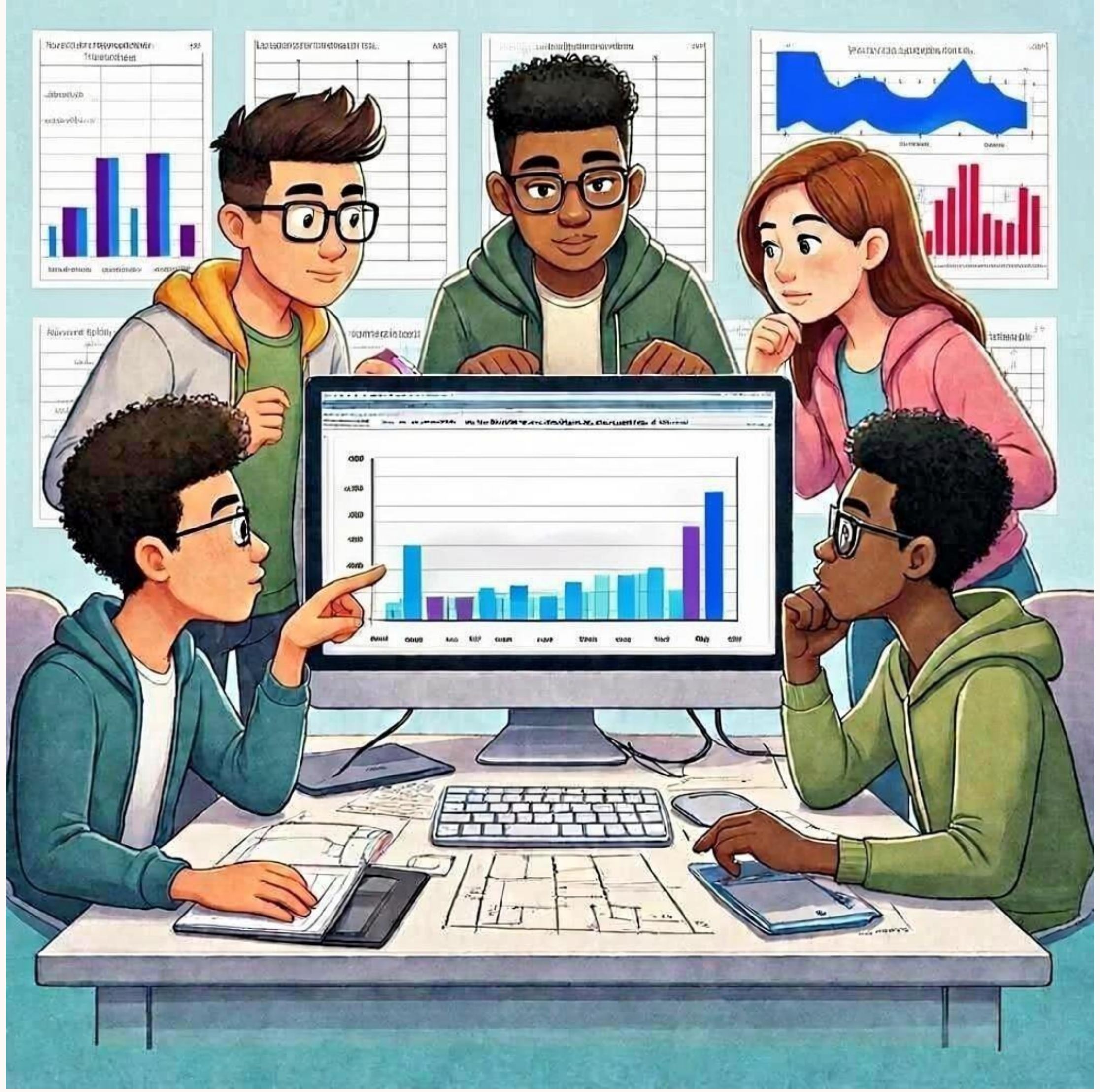
Environmental impact

Emission reduction, purification, resource recycling

Social impact

Creation of new jobs, improvement of the quality of life of the population

Target audience and market assessment



Who will use the results? (end users)

Potential partners and investors: Mining operators (Gold Fields, AngloGold Ashanti, Nordgold), national minerals commissions (Ghana Minerals Commission), industrial cybersecurity vendors (Claroty, Dragos), and academic research councils in Ghana and Russia.

Adaptation to the conditions of the region

How this framework is designed for African and Russian mining realities, not just laboratory conditions:

Climatic conditions

Ghana, DRC, and West African mining regions operate in hot, humid, high-dust environments. The edge deployment phase specifically targets low-cost hardware (Raspberry Pi class) that operates reliably in these conditions without climate-controlled server infrastructure.

Infrastructure

Many African mining sites have intermittent grid power and limited internet bandwidth. The BWOA feature selection stage reduces model size to enable offline or low-connectivity detection, and the pipeline is designed to run on edge hardware rather than requiring continuous cloud access.

The human factor

The output layer produces human-readable attack class labels rather than raw anomaly scores. Deployment roadmap includes a localization phase (Phase 3) for training local cybersecurity personnel at partner mining sites.

Solutions (as example): autonomy, protection, ease of repair, localization of production

The key question:

Why it works in the field: Phase 1 data collection is designed to happen AT real mining sites, meaning sites, meaning the model is trained on actual deployment-environment traffic from the start, not start, not retrofitted from a lab setting.

RUSSIAN-AFRICAN RAW MATERIALS DIALOGUE

Implementation plan and economics

01

Stage 1: Development and testing

The next 1-2 years: completion of R&D, prototyping, laboratory tests

02

Stage 2: Pilot implementation implementation

Real-world testing, feedback collection, revision

03

Stage 3: Scaling

Year 3 onwards: Scaled deployment across partner mining operators. Open-source release of adapted model and Ghanaian cloud traffic dataset. Expansion to additional African and Russian sites via forum network.

Budget estimate: Phase 1 data collection (AWS EC2 infrastructure, CICFlowMeter deployment, travel to pilot sites): approx. USD 8,000 to 15,000. Phase 2 model training (GPU compute): approx. USD 3,000 to 5,000. Phase 3 edge hardware and pilot deployment: approx. USD 5,000 to 10,000. Impact payback: Each prevented OT intrusion event avoids an estimated USD 50,000 to 500,000 in production downtime and equipment damage at a mid-scale mining operation. ROI is achievable within the first pilot deployment year.

Required partners: Mining operator with SCADA/IoT infrastructure (pilot site), GPU compute sponsor or academic HPC access, industrial cybersecurity dataset provider (SWaT/BATADAL access), and Russian-African academic co-investigators through this forum network.

Conclusion and invitation to dialogue

African and Russian mining digitalization is outpacing OT cybersecurity. This project delivers a validated, resource-conscious intrusion detection framework purpose-built for the digital mine.

Next: Phase 1 OT data collection; model retraining; peer-reviewed publication; expansion via Russian-African scientific network.

John Okyere | johnokyere.xyz | hello@johnokyere.xyz| +233591518686| University of Education, Winneba

Additional materials (optional)*

Mathematical calculations

Detailed graphics

List of the team's publications on the topic

Letters of support from potential customers (if any)

** This slide is not included in the main presentation, but it can be used to answer the jury's questions*